

Part C — Reflection Paper (Loan Approval Classification)

1. What did I implement?

In this assignment, I implemented a complete machine learning pipeline for **loan approval prediction** using a cleaned dataset. First, I reproduced the Lesson 5 preprocessing steps, including handling missing values, fixing inconsistent categorical values, encoding variables, scaling numerical features, and creating new engineered features such as **DebtToIncome** and **IncomePerYearEmployed**. These steps ensured that the dataset was clean and suitable for classification models.

After preprocessing, I split the dataset into training (80%) and testing (20%) sets using stratified sampling to maintain class balance. I then trained three classification models: **Logistic Regression** and **Random Forest** from the lesson, and an additional model, **Decision Tree**, which I researched independently. These models were used to predict whether a loan application should be approved (1) or rejected (0).

2. Comparison of Models

During the sanity check, I tested one sample from the dataset to compare predictions across models. The actual label was **Approved (1)**. Logistic Regression predicted **0 (Rejected)**, and Random Forest also predicted **0 (Rejected)**, while the Decision Tree correctly predicted **1 (Approved)**.

This shows that Logistic Regression and Random Forest failed to correctly classify this specific case, while Decision Tree gave the most realistic prediction. This is consistent with overall performance metrics, where Decision Tree achieved the best balance between accuracy and F1-score.

3. Understanding Random Forest

Random Forest is an ensemble learning method that builds multiple **decision trees** during training. Each tree makes a prediction, and the final output is determined by **majority voting** among all trees. This reduces overfitting and improves generalization compared to a single decision tree.

In this loan approval dataset, Random Forest worked well because it can handle non-linear relationships and feature interactions.

- **Advantage:** It is robust and reduces overfitting by averaging multiple trees.

- **Limitation:** It is less interpretable than a single decision tree and can be computationally heavier.

4. Other Algorithm (Decision Tree)

For the third classifier, I selected the **Decision Tree algorithm** because it is simple, interpretable, and widely used for classification problems like loan approval.

A Decision Tree works by splitting the dataset into branches based on feature values, creating a tree-like structure of decisions until it reaches a final prediction.

- **Advantage:** Easy to interpret and visualize, making it useful for understanding decision rules in loan approval systems.
- **Limitation:** It can overfit the training data if not properly controlled.

In my results, the Decision Tree achieved:

- Accuracy = **0.70**
- Precision = **0.706**
- Recall = **0.923**
- F1-Score = **0.80**

Compared to Logistic Regression and Random Forest, it performed best overall.

5. Metrics Discussion

The evaluation results showed different strengths for each model:

- **Logistic Regression** had the highest Recall (0.923), meaning it was very good at identifying approved loans, but it performed poorly in distinguishing rejected cases.
- **Random Forest** had better Precision (0.714), meaning it was more careful when predicting approvals, but its overall performance was moderate.
- **Decision Tree** achieved the best Accuracy (0.70) and the highest F1-score (0.80), showing the best balance between Precision and Recall.

This indicates that Logistic Regression is sensitive to identifying positives, Random Forest is more conservative, and Decision Tree provides a balanced performance for this dataset

6. Your Findings

Based on the results, I would choose the **Decision Tree model** for loan approval prediction in this dataset. It provided the most balanced performance across all evaluation metrics and

correctly predicted the test sample during the sanity check. Additionally, its interpretability makes it suitable for real-world financial decisions where explanations are important.

However, I also learned that model performance depends heavily on dataset size and quality. With a small dataset like this, results may vary, and more data or advanced tuning could improve performance further.